Transient Thermal Analysis of a Simplified Nuclear Reactor Using Coupled Point Kinetics and Temperature Feedback

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1 Introduction

1.1 Background: Nuclear Reactor Operation

Nuclear reactors generate thermal energy through fission reactions, in which a heavy atomic nucleus such as uranium-235 absorbs a neutron and undergoes fission into two lighter nuclei [1]. This process releases on the order of 200 MeV. together with two or three neutrons and gamma radiation. The resulting energy can be extracted and used for electricity generation.

The neutrons released after a fission reaction can induce more fission events, creating a self-sustaining chain reaction. The behaviour of the chain reaction is characterised by the effective neutron multiplication factor k_{eff} which describes ratio of the number of neutrons in one generation to the number of neutrons in the preceding generation, accounting for both absorption and leakage. A reactor is described as critical if $k_{eff} = 1$, meaning that there is a steady neutron population in constant conditions. If $k_{eff} < 1$ then the reactor is subcritical and the neutron population will decrease over time if there is no external neutron source. If $k_{eff} > 1$ then the reactor is supercritical and the neutron population tends to increase with time.

A very important distinction is the difference between prompt and delayed neutrons. Prompt neutrons are released almost immediately after the fission event (within 10^{-14}s) from the heavy nucleus. They have energies of around 2 MeV. There are also delayed fission neutrons, which are released by some unstable daughter nuclei, called delayed neutron precursors, when they undergo beta decay. In a small fraction of cases this leaves them with a high amount of excitation energy. If this energy is higher than the nucleus' binding energy the nucleus with emit a free neutron, with characteristic emission times ranging from a few milliseconds to tens of seconds after the original fission event. These neutrons have much lower energies (200-500 keV). An important factor to define here is delayed neutron fraction β which is the ratio of delayed neutrons to the total number of fission neutrons produced. Although this fraction is small (0.0065 for uranium-235), it has a large impact on the timescale over which the reactor power changes. Their delayed emission makes controlled power adjustments more manageable than a system of only prompt neutrons. As different precursor species have different decay constants the behaviour is commonly represented using many precursor groups. In this project both one-group and six-group approximations are investigated.

All the heat produced by the enriched uranium in the core needs to be transferred to prevent the core from melting. In a typical system heat is conducted from the fuel pellets through the cladding to the surrounding coolant. In most cases the coolant is typically high pressure water, which acts both as a coolant and a moderator. The high pressures prevent the water from boiling inside the reactor vessel. Heat from the core is transferred through the coolant to a seperate secondary water circuit within the steam generator, where steam is produced and goes on to be dried and then passes through turbines to generate electricity. The cooler high pressure water then returns to the core to be reheated.

Reactor operation also involves controlling the reaction. Control rods can absorb neutrons to control the neutron population. An important feature of nuclear reactors is negative temperature feedback, whereby a temperature increase leads to a reactivity decrease. This is due to two main reasons: Doppler broadening and thermal expansion. As temperatures rise, heavy non fissile nuclei like uranium-238 vibrate more vigorously and so their resonance peak for neutron capture broadens, meaning fewer neutrons are available for absorption by fissile uranium-235. Also as the temperature increases the water expands more, decreasing the density of water molecules. The neutrons collide less frequently with the molecules so slow down less, meaning they are less likely to thermalise and slow down enough to be absorbed.

Therefore the interaction between neutron kinetics and heat transfer is very important to understand reactor behaviour.

1.2 Motivation for Computational Modelling

A nuclear reactor is governed by several interacting physical processes operating over different timescales. Neutron generation influences reactor power whereas thermal conduction, heat capacities and coolant flow influence the temperature response. This coupling produces behaviour which cannot be analysed with steady-state analysis.

Computational modelling allows us to solve systems of coupled differential equations. Numerical integration techniques allow the time evolution of reactor power and temperatures to be calculated for given

initial conditions.

In this project a lumped parameter thermal model is coupled to point reactor kinetics equations to investigate the transient response of a simple system. The thermal model represents the fuel, cladding and coolant as three interacting thermal masses, while the kinetics describe the evolution of reactor power and delayed neutron precursor populations. A linear temperature dependent reactivity relationship is introduced to examine the effect of negative thermal feedback.

1.3 Project Objectives

The project is based on three questions.

1. **How does negative temperature reactivity feedback influence the response and stability of the system?** The model is used to investigate the effects of temperature-dependent reactivity on power evolution, peak fuel temperature, oscillatory behaviour and convergence towards equilibrium.
2. **Which parameters have the greatest influence on peak fuel temperature?** Parameter sensitivity studies are conducted by varying coolant mass flow rate, imposed reactivity and the temperature-feedback coefficient. Dimensionless sensitivity coefficients are calculated to quantify the relative influence of these parameters.
3. **How does the representation of delayed neutron precursors affect the transient behaviour?** A one-group point-kinetics model is compared with a six-group model to investigate differences in power evolution and thermal response.

To address these questions, the governing equations are implemented in Python and integrated using SciPy. The overall aim is to demonstrate how simplified computational models can be used to investigate coupled systems and to critically evaluate the assumptions, accuracy and limitations that influence their predictions.

2 Mathematical Model

A lumped thermal model is used for the reactor, meaning physical components such as fuel pellets, cladding and coolant channels are treated as discrete "lumps" with an average temperature and no spatial distribution. This was coupled to point-reactor kinetics, where both one and six group approximations were considered.

2.1 Thermal Model

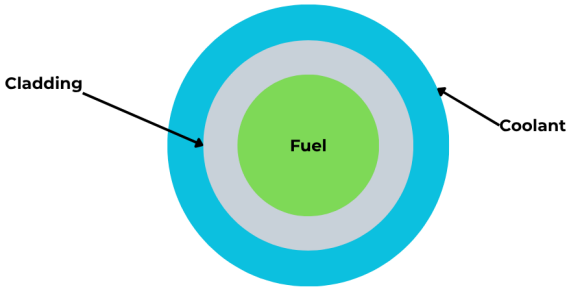


Figure 1: Simplified reactor model

The fuel undergoes fission and this provides the power for the reactor. In this model, as seen in Figure 1, the fuel is surrounded by cladding which isolates the fuel from the coolant (in this case water) and also holds the fuel cells in place. The coolant is used to transfer heat away from the reactor core and towards the steam generators.

The thermal system was represented by a reactor fuel, cladding and coolant, with respective temperatures T_f, T_c and T_w.

As well as the lumped parameter approximation, thermal properties such as heat capacities and thermal conductances were assumed constant. The power was assumed to be generated completely within the fuel.

The power generated within the reactor influences the temperature of the fuel, then the temperature of the cladding and then the coolant. We assume the input temperature of the coolant is fixed. The governing equations come from the conservation of energy and their derivation is below.

We represent the fuel as a mass with total heat capacity $C_f = m_f c_f$, where m_f is the fuel mass and c_f is the specific heat capacity. The rate of change of stored thermal energy is:

$$\frac{dU_f}{dt} = C_f \frac{dT_f}{dt}$$

There are two energy transfer processes: heat generation $P(t)$ and heat transfer from the fuel to cladding at rate $\dot{Q}_{fc}$. Applying conservation of energy:

$$\frac{dU_f}{dt} = P(t) - \dot{Q}_{fc}$$

We can approximate the fuel to cladding heat transfer using an effective thermal conductance K_{fc}.

$$\dot{Q}_{fc} = K_{fc}(T_f - T_c)$$

Substituting and rearranging gives:

$$C_f \frac{dT_f}{dt} = P(t) - K_{fc}\left(T_f - T_c\right)$$

Applying the same principle to the cladding, it gains energy from the fuel and loses it to the coolant across the cladding-coolant interface with conductance K_{cw}, giving us:

$$C_c \frac{dT_c}{dt} = K_{fc}\left(T_f - T_c\right) - K_{cw}\left(T_c - T_w\right)$$

The coolant gains energy from the cladding and transports it out of the system by mass flow. Assuming constant heat capacity the power lost from the coolant is $\dot{m}c_p\left(T_w - T_{in}\right)$, where T_{in} is the inlet temperature of the coolant and c_p is its specific heat capacity. This gives us the governing heat transfer equations:

$$\begin{aligned}
C_f \frac{dT_f}{dt} &= P(t) - K_{fc}\left(T_f - T_c\right) \\
C_c \frac{dT_c}{dt} &= K_{fc}\left(T_f - T_c\right) - K_{cw}\left(T_c - T_w\right) \\
C_w \frac{dT_w}{dt} &= K_{cw}\left(T_c - T_w\right) - \dot{m}c_p\left(T_w - T_{in}\right)
\end{aligned}$$

These are used to model the temperature variation throughout reactor.

2.2 Point Reactor Kinetics

Initially we assumed a constant power but then introduced the point reactor kinetics equations. These equations describe how the power will change according to the variables of the system.

In this approximation the spatial distribution of the neutron flux is assumed to be fixed while the amplitude varies with time. This allows the neutron population to be represented by a single variable $n(t)$.

Reactivity is defined in terms of k_{eff} as:

$$\rho = \frac{k_{eff} - 1}{k_{eff}}$$

A reactor is critical when $\rho = 0$ ($k_{eff} = 1$).

Positive reactivity leads to a neutron population increase and negative reactivity leads to a neutron population decrease.

The total effective delayed neutron fraction is:

$$\beta = \sum_{i=1}^{N} \beta_i$$

where β_i is the effective delayed neutron fraction associated with group i.

Within the point-kinetics approximations the neutron population balance is [2]:

$$\frac{dn}{dt} = \frac{\rho - \beta}{\Lambda} n + \sum_{i=1}^{N} \lambda_i C_i$$

where Λ is the prompt neutron generation time, λ_i is decay constant of group i and C_i is its effective precursor state.

The first term represents the net prompt contribution to neutron population evolution, while the sum accounts for neutrons originating from delayed-neutron precursor decay.

Delayed neutron precursors are produced during fission and subsequently undergo radioactive decay. For precursor group i the rate of change of precursor population is modelled as the difference between production and decay.

In the point kinetics approximation the production term is proportional to neutron population and the decay rate is proportional to the rate at which delayed neutrons are released.

$$production\ rate = \frac{\beta_i}{\Lambda} n$$

$$decay\ rate = \lambda_i C_i$$

This leads to:

$$\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} n - \lambda_i C_i$$

To formulate this in terms of power we used the assumption that:

$$P(t) = k_P n(t)$$

where k_P is a constant proportionality factor. Differentiating with respect to time gives:

$$\frac{dP}{dt} = k_P \frac{dn}{dt}$$

and defined power scaled precursor states:

$$C_{P,i} = k_P C_i$$

This leads to:

$$\frac{dP}{dt} = \frac{\rho - \beta}{\Lambda} P + \sum_{i=1}^{N} \lambda_i C_{P,i}$$

$$\frac{dC_{P,i}}{dt} = \frac{\beta_i}{\Lambda} P - \lambda_i C_{P,i}$$

2.3 Negative Temperature Feedback

A widely used model for reactivity as a function of fuel temperature relies on assumption that the fuel temperature coefficient of reactivity α is inversely proportional to the square root of absolute temperature [3] so:

$$\alpha\left(T_f\right) = \frac{d\rho}{dT_f}$$

Where:

$$\alpha\left(T_f\right) \propto \ -\frac{1}{\sqrt{T_f}}$$

However, for small deviations around the equilibrium temperature we can simply Taylor expand around the equilibrium temperature $T_{f,0}$ which gives us:

$$\rho(T_f) \approx \rho(T_{f,0}) + \left.\frac{d\rho}{dT_f}\right|_{T_{f,0}} (T_f - T_{f,0}).$$

And defining $\alpha = \frac{d\rho}{dT_f}|_{T_{f,0}}$ gives us:

$$\rho \approx \rho_0 + \alpha\left(T_f - T_{f,0}\right)$$

For negative feedback $\alpha < 0$.

2.4 Initial and Equilibrium Conditions

To investigate the effect of disturbances from equilibrium the system was started in a steady state determined by the reference power P_0.

At equilibrium $\frac{dT}{dt} = 0$ for all temperatures so setting all time derivatives to zero in the thermal equations gives:

$$\begin{aligned}
0 &= P(t) - K_{fc}\left(T_f - T_c\right) \\
0 &= K_{fc}\left(T_f - T_c\right) - K_{cw}\left(T_c - T_w\right) \\
0 &= K_{cw}\left(T_c - T_w\right) - \dot{m}c_p\left(T_w - T_{in}\right)
\end{aligned}$$

Which can be solved to get the initial temperatures of the fuel, coolant and cladding.

$$T_{w,0} = T_{in} + \frac{P_0}{\dot{m}c_p}$$

$$T_{c,0} = T_{w,0} + \frac{P_0}{K_{cw}}$$

$$T_{f,0} = T_{c,0} + \frac{P_0}{K_{fc}}$$

The delayed neutron precursor states were also initialised at equilibrium by setting $\frac{dC_{P,i}}{dt} = 0$

This gives:

$$C_{P,i} = \frac{\beta_i}{\Lambda\lambda_i} P$$

Temperature feedback equilibrium:

At steady state power $\rho = 0$. So putting this into the linear feedback model gives:

$$\rho\left(T_{f,eq}\right) = 0$$

$$T_{f,eq} = T_{f,0} - \frac{\rho_0}{\alpha}$$

And if α is negative and ρ_0 is positive the fuel equilibrium temperature will be greater than its initial, as expected.

Equilibrium power:

This can be obtained from the steady state thermal equations in terms of the equilibrium fuel temperature and coolant inlet temperature as:

$$P_{eq} = \frac{T_{f,eq} - T_{in}}{\frac{1}{\dot{m}c_p} + \frac{1}{K_{fc}} + \frac{1}{K_{cw}}}$$

3 Numerical Methodology

The coupled thermal and point kinetics equations derived in section 2 were implemented numerically in Python. Numerical integration was performed using solve_ivp function from the SciPy library.

The computational implementation was done progressively in increasing complexity, beginning with the three node thermal model, before introducing point kinetics, temperature dependent feedback and then the six delayed neutron precursor groups.

The governing equations were expressed in the general form:

$$\frac{dy}{dt} = f(t, y, \theta)$$

Where initially $y(0) = \ y_0$

And y_0 is the vector of time dependent state variables and θ represents the model parameters.

In the case of the one-group kinetics model y was defined as:

$$y(t) = \ [T_f \ T_c \ T_w \ P \ C_P]$$

And in the six-group model:

$$y(t) = \ [T_f \ T_c \ T_w \ P \ C_{P,1} \ \cdots \ C_{P,6}]$$

At each evaluation the numerical solver supplied the current state vector to the model function which calculates the instantaneous derivatives of all variables using the equations in section 2. The derivatives are returned to the solver to allow the integration to proceed.

3.1 Numerical Integration

Using SciPy's solve_ivp[4] provides multiple ways of solving systems of equations. The majority of the investigation was done using the adaptive Runge-Kutta method RK45. This method is improved over the standard RK4 method by adjusting the integration step size to satisfy relative and absolute error tolerances.

The introduction of point reactor kinetics produced a system with variables varying over different time scales. For example the prompt neutron generation time is much shorter than the thermal response times of the reactor components. These timescale differences can produce numerical stiffness, making it hard for numerical integration methods to work well.

For the coupled six group model the implicit backward differentiation formula (BDF) method was therefore used as BDF methods are more suitable for stiff systems.

Solver accuracy was controlled using relative and absolute tolerances, with representative settings of

$$\begin{aligned} rtol &= 10^{-8} \\ atol &= 10^{-10} \end{aligned}$$

The relative tolerance controls the permitted local error in proportion to the value of the solution variables, while the absolute tolerance provides error scale for variables close to zero.

3.2 Initial Conditions and Numerical Verification

Initial conditions were calculated analytically from the relationships in section 2.4.

The physical and numerical parameters were stored in a Python dataclass and the initial conditions were calculated from the parameters within Python, to reduce risk of inconsistencies when variables were modified. The parameters used were illustrative and not calibrated to a specific reactor.

Some numerical verification was done to assess whether the numerical implementation accurately solved the coupled equations.

An initial verification test was conducted using a simplified single node thermal model with constant power and fixed surrounding temperature. Under these conditions the solution to the equation is an analytic exponential solution. The numerical solution was compared to this.

The model was also tested under steady state conditions and when initialised using equilibrium temperatures and zero reactivity, the calculated temperatures and power were expected to stay constant.

3.3 Parameter Sensitivity Analysis

A parameter sensitivity analysis was conducted to investigate the influence of selected parameters on thermal response.

The parameters considered were: coolant mass flow rate ($\dot{m}$), imposed reactivity (ρ_0) and temperature-reactivity feedback coefficient (α).

For each parameter a set of simulations was performed while the remaining parameters were held constant. All simulations started from a common reference initial state.

The quantity recorded was the maximum fuel temperature $T_{f,max}$.

To quantify the influence of each parameter a sensitivity coefficient was calculated.

To ensure this result is valid regardless of the units this coefficient needs to be dimensionless. So if we define this coefficient as the ratio of relative change in output (T) to the relative change in input (x).

$$\textit{Relative change in } x = \frac{\Delta x}{x}$$

$$\textit{Relative change in } T = \frac{\Delta T}{T}$$

Then the sensitivity coefficient equals:

$$S_{T,x} = \frac{x}{T}\frac{\Delta T}{\Delta x}$$

And taking the limit as Δx goes to 0:

$$S_{T,x} = \lim_{\Delta x \to 0}\left(\frac{x}{T}\frac{\Delta T}{\Delta x}\right) = \frac{x}{T}\frac{\partial T}{\partial x}$$

So the coefficient:

$$S_{T,x} = \frac{x}{T_{f,max}(x_0)}\left.\frac{\partial T_{f,max}}{\partial x}\right|_{x=x_0}$$

Where x_0 denotes the value of the parameter under investigation.

To approximate the partial derivative a central finite difference was used

$$\left.\frac{\partial T_{f,max}}{\partial x}\right|_{x=x_0} = \frac{T_{f,max}(x_0+\Delta x) - T_{f,max}(x_0-\Delta x)}{2\Delta x}$$

The sensitivity calculations were repeated with smaller perturbation sizes to assess convergence, allowing a comparison of factors with different units.

3.4 Comparing One and Six-Group Kinetics

To investigate the influence of delayed neutron representation, simulations were performed using both one-group and six-group kinetics formulations. The thermal parameters, reactor power, imposed reactivity and temperature feedback coefficient were kept the same between the models.

The total delayed-neutron fraction was also matched:

$$\beta = \sum_{i=1}^{6}\beta_i$$

The one-group model and six-group model should have the same total equilibrium precursor inventory.

For the one group model:

$$C_0 = \frac{\beta P_0}{\Lambda\lambda_{eff}}$$

Where C_0 is the equilibrium precursor inventory.

For the six group model:

$$C_{i,0} = \frac{\beta_i P_0}{\Lambda\lambda_i}$$

$$C_{total,0} = \sum_{i=1}^{6} C_{i,0} = \frac{P_0}{\Lambda}\sum_{i=1}^{6}\frac{\beta_i}{\lambda_i}$$

Now equating these gives:

$$\lambda_{eff} = \frac{\beta}{\sum_{i=1}^{6}\frac{\beta_i}{\lambda_i}}$$

This guarantees that the simplified model preserves the same total precursor inventory.

These all provide a framework for investigating the coupled reactor model and evaluating the influence of modelling assumptions.

3.5 Reference Parameters

Parameter	Symbol	Reference value	Units
Thermal power	P_0	10^6	W
Fuel thermal capacity	C_f	5×10^5	J/K
Cladding thermal capacity	C_c	10^5	J/K
Coolant thermal capacity	C_w	2×10^5	J/K
Fuel–cladding conductance	K_{fc}	10^4	W/K
Cladding–coolant conductance	K_{cw}	2×10^4	W/K
Coolant mass flow	$\dot{m}$	10	kg/s
Coolant specific heat	c_p	4200	J/(kg K)
Coolant inlet temperature	T_{in}	300	K
Imposed reactivity	ρ_0	0.001	-
Temperature feedback coefficient	α	-10^{-5}	1/K

4 Numerical Verification

Numerical verification was undertaken to assess the accuracy of the computational implementation developed in Section 3.

4.1 Verification Against Analytical Solution

This test considered a single node thermal system for which an exact analytical solution can be derived.

The fuel temperature was assumed to evolve under constant thermal power, fuel heat capacity and a constant cladding temperature and fuel to cladding conductance.

The governing energy balance:

$$C_f \frac{dT_f}{dt} = P - K_{fc}\,(T_f - T_c)$$

And from before:

$$T_{eq} = T_c \; + \; \frac{P}{K_{fc}}$$

So the governing equation can be rewritten as:

$$\frac{dT_f}{dt} = -\frac{K_{fc}}{C_f}\,(T_f - T_{eq})$$

Which can be integrated to give:

$$T_f(t) = T_{eq} \; + \; \; [T_f(0) - T_{eq}] \; \exp(-\frac{K_{fc}}{C_f}t)$$

The absolute numerical error was calculated as:

$$\varepsilon(t) = \; |T_f(t) - T_{f,analytic}(t)|$$

The measured maximum absolute error was $7.9 \times 10^{-9} K$, so the numerical solution agreed with the analytical solution to within the measured error.

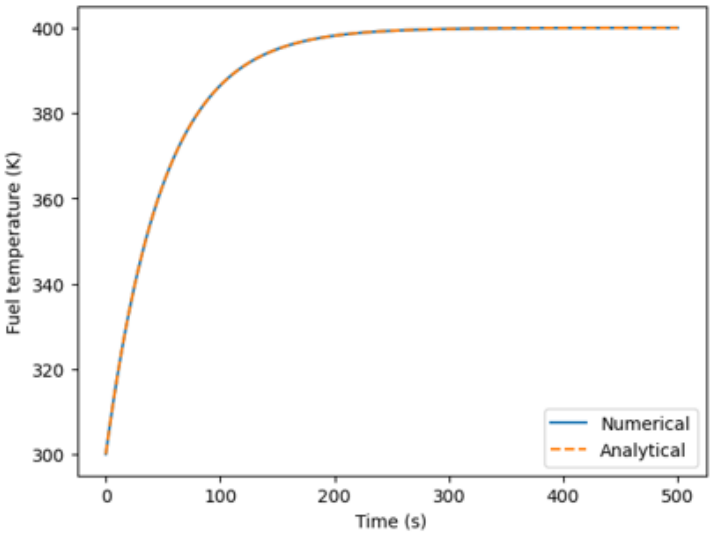


Figure 2: Numerical vs Analytic fuel temperature evolution

4.2 Steady-state Equilibrium

A second test was performed by starting the system in the equilibrium state, given by the equations in section 2.4.

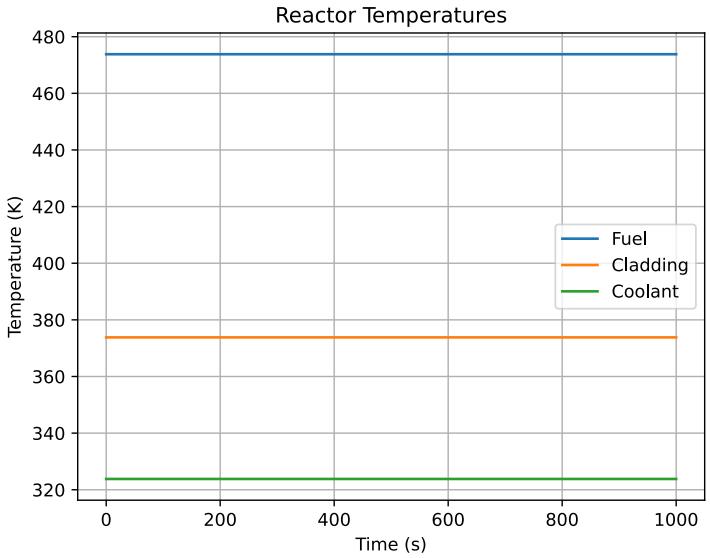


Figure 3: One group temperature evolution with no imposed reactivity

With zero imposed reactivity Figure 3 shows that the system remains in this equilibrium state. This was also repeated for the six group model:

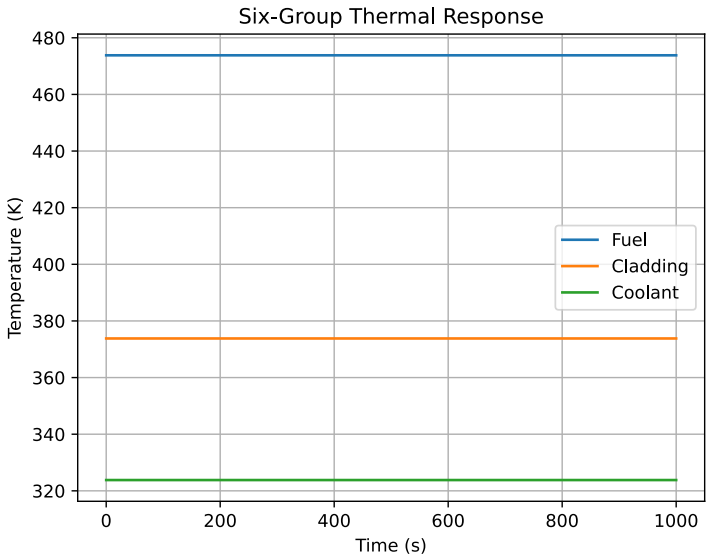


Figure 4: Six group temperature evolution with no imposed reactivity

4.3 Equilibrium Under Temperature Feedback

An additional check was possible for the model with negative temperature reactivity.

At a steady state reactor power, the kinetic equations require zero reactivity ($\rho = 0$)

From section 2.4, the equilibrium fuel temperature is:

$$T_{f,eq} = T_{f,0} - \frac{\rho_0}{\alpha}$$

Using illustrative values of $\rho_0 = 0.001$[5] and $\alpha = -10^{-5}$, the predicted equilibrium temperature is

$$T_{f,eq} = 573.8\ K$$

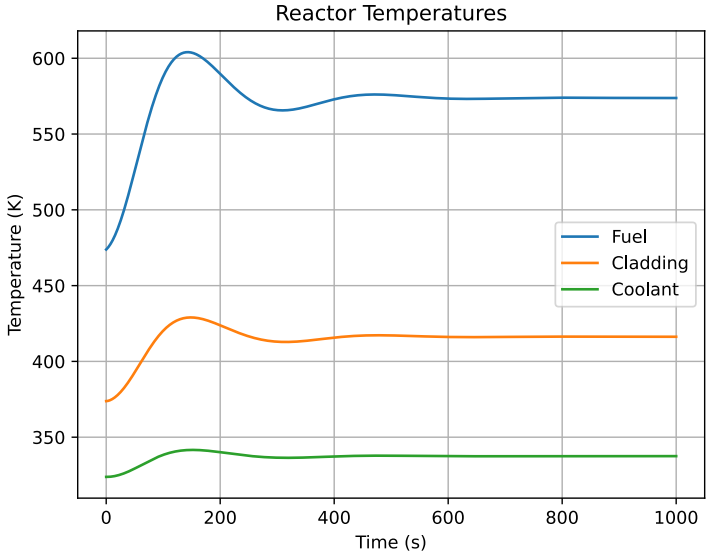


Figure 5: One-group temperature evolution for $\rho_0 = 0.001$

As can be seen from Figure 5 , the equilibrium fuel temperature aligns with the predicted value.

5 Results and Discussion

5.1 Transient Behaviour Without Temperature Feedback

The initial investigation considered the coupled thermal and point-kinetics model without temperature dependent feedback. A constant reactivity insertion of $\rho = 0.001$ was imposed at the beginning of the simulation.

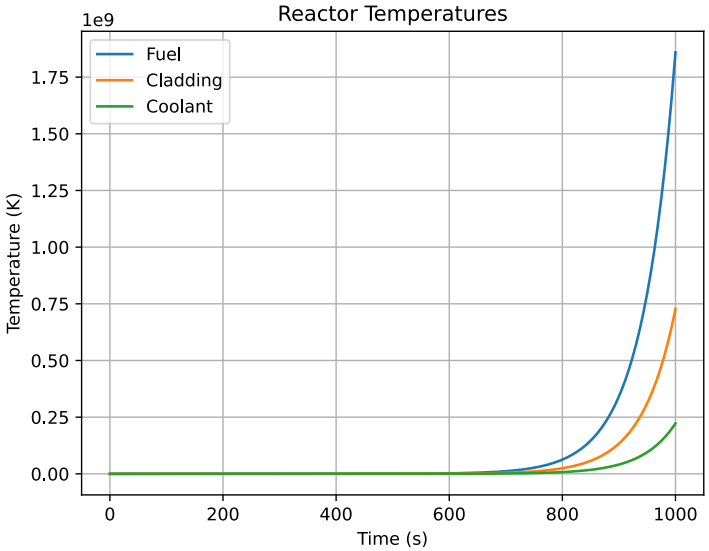


Figure 6: Temperature evolution with no feedback

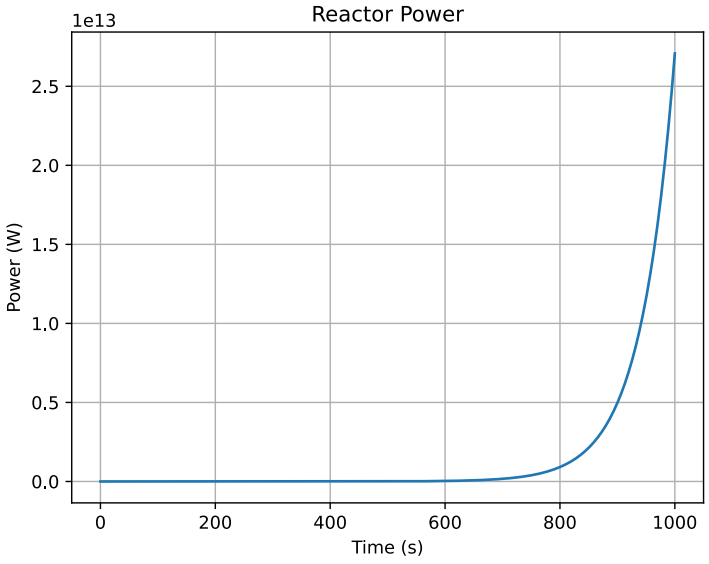


Figure 7: Power evolution with no feedback

The simulation produced an increasing reactor power, along with rising fuel, cladding and coolant temperatures. The fuel temperature responded most directly to the increased heat generation, while the cladding and coolant had some delay due to the finite heat capacities and thermal conductances.

This basic model demonstrated the correct relationship, that an increased neutron population leads to an increased reactor power.

However this model is very unrealistic as power exponentially grows and leads to a final fuel temperature

of greater than 10^9 K as well as power exceeding 10^{13} W. This demonstrates the key limitation of this model and the importance of negative feedback within a reactor to oppose increases in power.

This motivated the introduction of feedback within the investigation.

5.2 Influence of Negative Temperature Reactivity Feedback

As mentioned in section 2.3 the reactivity was modelled as

$$\rho \;=\; \rho_0 + \alpha(T_f - T_{f,0})$$

The illustrative parameters used were $\rho = 0.001$ and $\alpha = \; -10^{-5}\; K^{-1}$

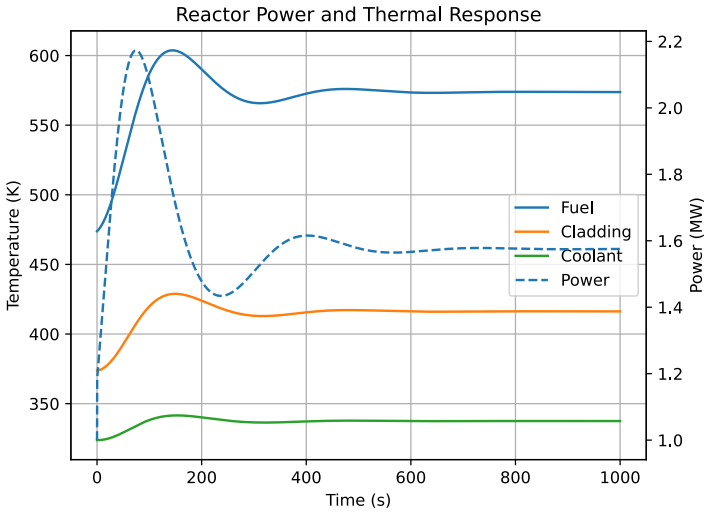


Figure 8: Reactor Power and Thermal Response with Feedback

In contrast to the previous simulation the introduction of feedback produced a bounded solution that approached a new equilibrium. The response was characteristic of an underdamped response, with oscillations decreasing in amplitude before approaching a steady value. The oscillations arise from the interaction between the reactor kinetics, feedback and thermal inertia. Any increase in power produces an increase in fuel temperature, which reduced the reactivity through the feedback. This opposes further power growth and leads to power oscillating.

As the reactor components have finite heat capacities, a delay can be seen between the power and the thermal response.

5.3 Sensitivity to Coolant Flow Rate

Here a parameter scan is conducted to graphically analyse the effects of variation of various parameters. The influence of coolant flow rate on the maximum fuel temperature was investigated by varying $\dot{m}$ while keeping the remaining parameters fixed.

The parameter scan used the following flow rates:

$$\dot{m} = 5,\; 7.5,\; 10,\; 12.5,\; 15\; kg\; s^{-1}$$

The coolant mass flow rate acts through the term

$$\dot{m} c_p \left(T_w - T_{in}\right)$$

The coupled system also contains feedback so a change in the coolant flow affects the fuel temperature, which alters reactivity and subsequently changes reactor power meaning the relationship is dependent on both thermal and kinetic response.

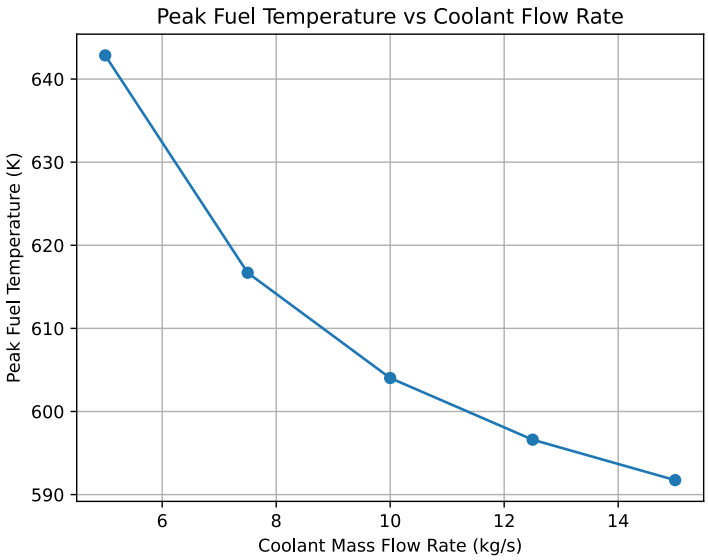


Figure 9: Peak fuel temperature vs coolant flow rate

The trend shows a monotonically decreasing function. This result is expected as moving more coolant through the system should reduce the temperature of all components as heat generated is transferred away faster.

5.4 Sensitivity to Imposed Reactivity

The effect of imposed reactivity was investigated by varying ρ_0. For positive reactivity insertions the neutron population tends to increase, causing power to increase.

The parameter scan used the following reactivities:

$$\rho_0 = 0.0002,\ 0.0005,\ 0.001,\ 0.0015,\ 0.002$$

The analytical relationship predicts for

$$T_{f,eq} = T_{f,0} - \frac{\rho_0}{\alpha}$$

And differentiating with respect to ρ_0 gives

$$-\frac{1}{\alpha}$$

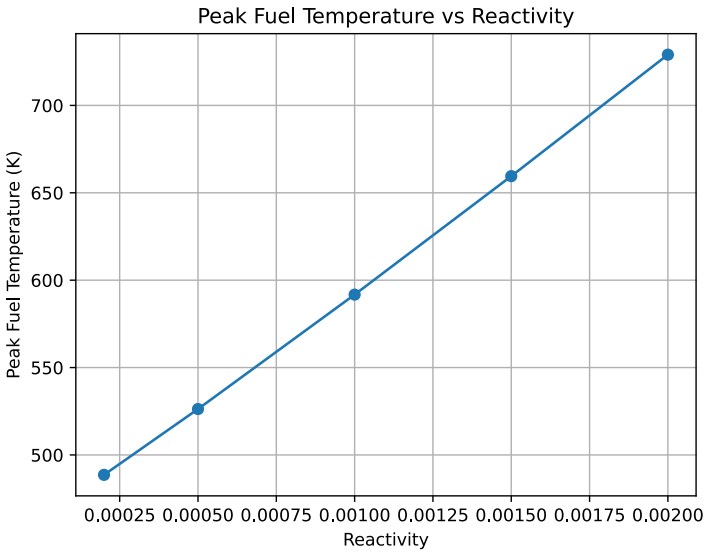


Figure 10: Peak fuel temperature vs reactivity

This means the predicted relationship is linear between imposed reactivity and equilibrium fuel temperature. What is plotted in Figure 10 are the peak fuel temperatures, which do not pbey this exact relationship. What we can see from the figure is that the relationship is still very close to linear and has an R^2 value of 0.99989.

5.5 Sensitivity to Temperature Feedback Coefficient

For negative feedback, making the coefficient more negative leads to a larger drop in reactivity per kelvin of temperature rise.

The parameter scan used the following coefficients:

$$\alpha = \{-2, -1.5, -1, -0.5, -0.25\} \times 10^{-5} \text{ K}^{-1}$$

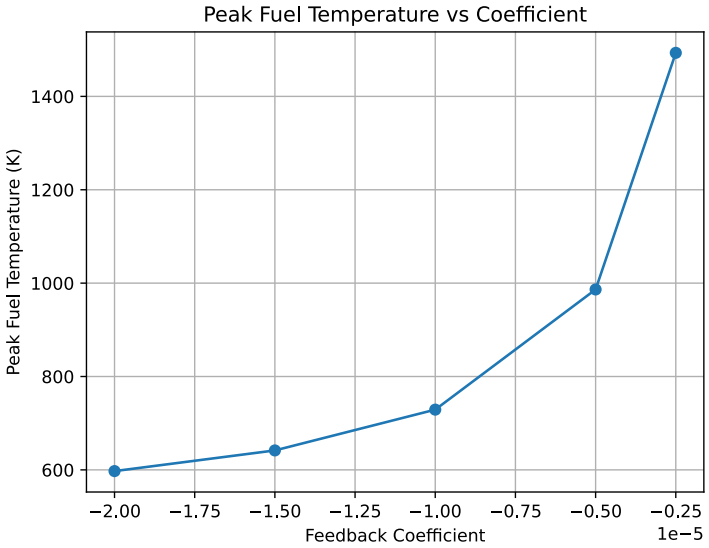


Figure 11: Peak fuel temperature vs feedback coefficient

The analytical equilibrium relationship provides an indication of the effects of changing α.

$$T_{f,0} - \frac{\rho_0}{\alpha}$$

So for more negative alpha the equilibrium fuel temperature is expected to be lower. We do not predict a linear trend, differentiating with respect to alpha gives:

$$\frac{\rho_0}{\alpha^2}$$

So as α approaches 0 (increasing α) we expect the equilibrium fuel temperature to increase rapidly.

We can see this trend is held by the peak fuel temperature too, in Figure 11, it increases rapidly with α.

5.6 Dimensionless Sensitivity Analysis

To compare the relative influence of parameters with different physical units, dimensionless sensitivity coefficients were used.

Recalling the sensitivity of maximum fuel temperature was defined as:

$$S_{T,x} = \frac{x}{T_{f,max}(x_0)} \left.\frac{\partial T_{f,max}}{\partial x}\right|_{x=x_0}$$

Where x_0 denotes the value of the parameter under investigation.

The calculated coefficients are:

Parameter	Reference Value	Sensitivity
Coolant flow rate $\dot{m}$	10 kg/s	-0.0691
Imposed Reactivity	0.001	0.275
Temperature feedback coefficient α	$-10^{-5}K^{-1}$	-0.266

From this data we can see the imposed reactivity had the largest local sensitivity among the parameters tested, meaning any change in the reactivity had the largest impact on the peak fuel temperature.

As the sensitivity coefficients are evaluated locally around the reference parameter set the values of the coefficients may change for different conditions or larger perturbations.

6 Comparison of One-group and Six-group Point Kinetics

The representation of delayed-neutron precursors is an important consideration in point reactor kinetics modelling. Although delayed neutrons constitute a small fraction of the total neutron population generated through fission they are very important for influencing reactor power.

In the one group approximation all delayed neutron precursors are represented by a single effective state C with delayed neutron fraction β and decay constant λ_{eff}.

The six group approximation uses six different precursor states, each with its own decay time $\tau_i = \frac{1}{\lambda_i}$

From Section 3.4 the effective decay constant can also be written as

$$\frac{1}{\lambda_{eff}} = \sum_{i=1}^{6} \frac{\beta_i}{\beta} \frac{1}{\lambda_i}$$

This provides a different way of interpreting the effective precursor lifetime. It is a weighted average of the individual lifetimes with weights $w_i = \frac{\beta_i}{\beta}$. This means that groups contributing a larger fraction of delayed neutrons receive more weight in the effective lifetime.

6.1 Selection of an Effective Decay Constant

In order to meaningfully compare the two models we need consistent parameters.

As was done in section 3 we can see that both the effective delayed neutron decay fraction and the effective decay constant are matched.

Group	Delayed Neutron Fraction β_i	Decay Constant λ_i
1	0.0005	0.02
2	0.0010	0.05
3	0.0015	0.10
4	0.0015	0.20
5	0.001	0.50
6	0.001	1.00

Which give $\beta = 0.0065$ and λ_{eff} of approximately 0.0922 s^{-1}

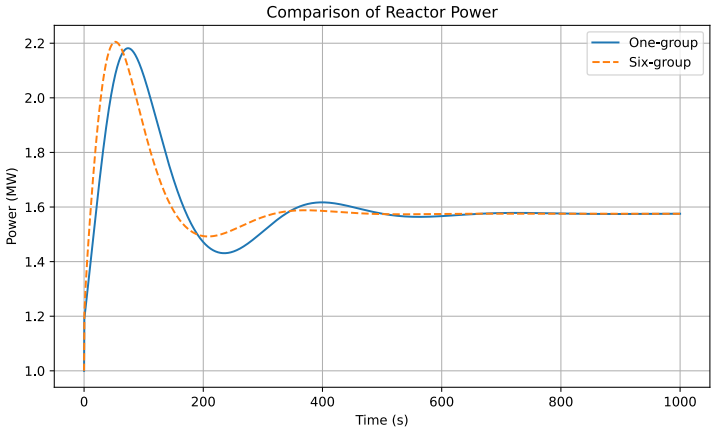


Figure 12: Comparison of Reactor Power

Figure 12 compares how the reactor power varies between the two models. Figure 12 shows the six-group model produced a higher peak power and the peak occurred earlier in time.

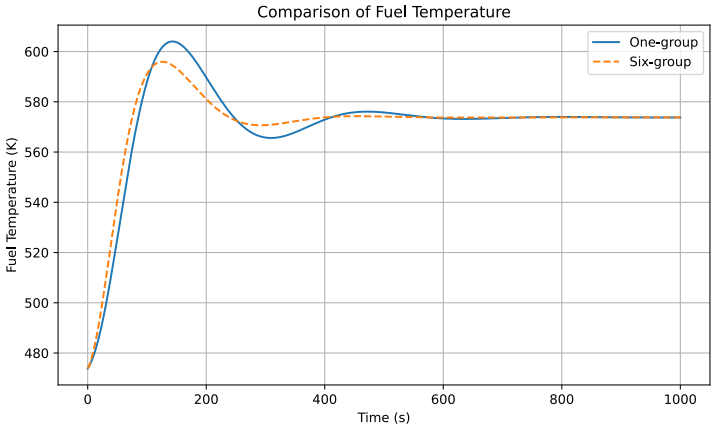


Figure 13: Comparison of Fuel Temperature

Although the six-group model predicts a higher maximum instantaneous power, the corresponding fuel temperature peak is lower. This is consistent with differences in duration and timing of the power, since the temperature depends on the history of heat generation and not instantaneous power alone.

6.2 Comparison of Equilibrium Behaviour

We can see that both models reach the same equilibrium fuel temperature.

From the feedback equation in Section 2.4 we can see that the equilibrium fuel temperature is independent of the number of precursor groups.

7 Model Limitations

Although the computing model developed reproduces some important features of coupled reactor kinetics its predictions are subject to limitations arising from simplifying assumptions made.

The principal limitations made concern the lumped thermal approximation, simplified neutron kinetics, representation of temperature feedback and selection of parameters.

7.1 Thermal and Cooling Model Limitations

The thermal model represents the reactor components as three discrete masses, each with a uniform temperature. In a physical reactor there are spatial temperature variations throughout the fuel, cladding and coolant. For example, within a fuel rod heat generated by fission produces radial and axial temperature gradients and the coolant has temperature variations as it flows through the core.

As the model provides an average temperature, the temperature within parts of a reactor can be far higher than the average and so this needs to be considered when designing reactor components.

Furthermore heat transfer is represented using constant conductances K_{fc} and K_{cw}, whereas in reality these characteristics depend on factors including temperature, geometry and coolant flow conditions. The model also assumes constant heat capacities for the reactor components, which in reality are affected by temperature.

The model also neglects spatial variation in coolant velocity, pressure and temperature.

A possible extension would be to replace the single fuel temperature-node with one dimensional radial heat conduction model, providing a more detailed representation of heat transfer or include temperature dependence on fluid flow.

7.2 Neutron Kinetics and Feedback Limitations

The neutron kinetics model is based on point reactor approximation, where the spatial distribution of the neutron flux is unchanging and only the amplitude varies with time. This reduces the computational complexity associated with spatially resolved neutron transport as it keeps the equations ordinary differential equations. However the flux of neutrons does vary spatially and leads to important effects such as spatial xenon oscillations.

The six-group model represents additional precursor groups but it still has the same limitations mentioned.

The temperature dependent feedback was represented using a linear relationship, allowing the influence of negative feedback to be investigated, but it does not capture the true behaviour. In reality reactivity varies on many different factors, including fuel temperature, moderator temperature and density. This assumption is also not valid for large temperature changes where we cannot assume a constant feedback coefficient.

The model also ignores any control systems, such as control rods or any other mechanisms.

7.3 Parameter Selection and Numerical Uncertainty

The parameters selected were primarily for illustrative purposes and do not reflect any specific reactor design. Therefore the timescales, peak powers and temperatures are not realistic interpretations of an operational nuclear reactor.

The coupled system has processes operating over vastly different timescales (such as prompt neutron generation and thermal energy transfer) and such differences can produce numerical stiffness. The use of an implicit BDF solver reduces some of the difficulties associated with stiff systems.

The sensitivity analysis is also subject to limitation as the dimensionless sensitivity coefficients were calculated using finite difference approximations. These coefficients therefore describe local sensitivity rather than behaviour of model across entire parameter space. Furthermore varying just one parameter at a time does not capture any effects from changing two or more parameters at the same time.

In this case a more comprehensive analysis would involve simultaneous variation of multiple parameters across a broader parameter range.

8 Conclusions

The investigation aimed to develop, verify and analyse a model of the coupled thermal and neutron kinetic behaviour of a simplified nuclear reactor.

The governing equations were implemented in Python and numerically integrated using SciPy. The model was developed incrementally, beginning with a simplified thermal system before incorporating delayed neutron precursor dynamics, negative temperature feedback and a six group point kinetics formulation. Numerical verification was also done to establish confidence in the solutions of the equations.

The introduction of negative feedback significantly influenced the response of the system. Without feedback, a sustained positive reactivity insertion produced continuously increasing reactor power and temperatures. After including feedback the result was a bounded, damped oscillatory response. The fuel temperature approached an equilibrium of approximately 574K, consistent with the analytical prediction.

A sensitivity analysis was done to investigate the influence of coolant mass flow rate, imposed reactivity and temperature feedback coefficient on the maximum fuel temperature. Dimensionless coefficients were calculated and this demonstrated how parameter variation can be used to identify parameters that warrant further investigation.

The comparison between the one and six group model demonstrated that the representation of delayed neutron precursors can affect predicted transient behaviour. In particular, the simulations demonstrated that a higher maximum instantaneous reactor power does not necessarily correspond to a higher maximum fuel temperature, since the thermal response depends on the history of heat generation. Both models are subject to the limitations of the lumped parameter approximation, simplified temperature feedback and parameter selection.

Several extensions are possible to improve the work in the future. These include introducing spatial heat conduction within the fuel, incorporating temperature dependent properties, investigating multiple feedback mechanisms and performing better sensitivity analysis.

Overall the project demonstrates the application of computational physics to a real problem and allows the study of a simple reactor system.

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